

RAJAN KUMAR SINGH

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EDUCATION

Vellore Institute of Technology (VIT)

India

Bachelor of Technology in Computer Science & Engineering ; CGPA: 8.30/10

Aug 2022 – Jul 2026

Relevant Coursework : Data Structures & Algorithms, Operating Systems, DBMS

Computer Networks, Object-Oriented Programming, Software Engineering

SKILLS

Programming Languages	Go, C++, Javascript
Frameworks/Libraries	gorilla/mux, JWT
Backend Development	REST APIs, Authentication, TCP Networking, SDLC
Databases	SQL
Cloud / DevOps	AWS (EC2, S3, RDS), Docker, Git
Tools / Concepts	Linux (CLI), Concurrency (Goroutines/Channels), File I/O, Debugging(curl)

PROFESSIONAL EXPERIENCE

Ethara AI — Data Annotator

Mar 2026 – Apr 2026

- Evaluated and annotated **multimodal AI outputs** including images and videos based on prompt-following, visual quality, and AI-generation criteria
- Performed structured quality assessment by identifying inconsistencies, artifacts, and instruction-following errors across generated content
- Worked with guideline-driven annotation workflows requiring high attention to detail, consistency, and accurate observational reporting

PROJECTS

ByteHaven — Secure File Vault Backend | GitHub

July 2025 – Sept 2025

Go, PostgreSQL, JWT, Docker, MinIO (S3-Compatible)

- Developed **REST APIs** in **Go** with **JWT authentication** for secure upload/download, metadata listing, file deletion, and access-controlled sharing workflows
- Designed database schema with 5 relational tables in **PostgreSQL** (User, File, UserFile, Folder, Share)
- Implemented **SHA-256** based file deduplication with reference counting, preventing duplicate object storage
- Added security and reliability controls including MIME-type validation, upload size limit (10 MB), per-user quota enforcement, and rate limiting (2 req/min)

Caching Proxy Server (C++) | GitHub

Jan 2026 – Mar 2026

C++-based proxy server with response caching and request forwarding

- Built a caching proxy server in C++ that intercepts client requests, forwards them to an origin server, and returns responses.
- Implemented an in-memory caching layer using **unordered_map** to store and reuse responses, reducing redundant backend calls
- Parsed incoming HTTP requests to extract endpoints and dynamically construct origin URLs for request forwarding

OPEN-SOURCE CONTRIBUTIONS

Grafana(k6) : Contributed **pressSequentially** feature to the k6 load-testing tool, improving sequential keypress simulation; contribution merged into main branch